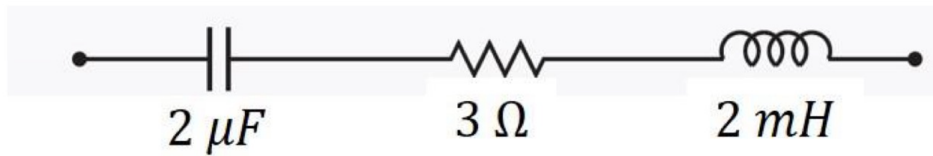


Assignment 3

Due Monday, Feb 9, 2026

- 1) For the RLC circuit below, compute the resonance frequency and quality factor. Also compute the impedance of the circuit at its resonance frequency.



a.)

$\omega_0, Z_{eq.} = R, V_0 = V_R = IR: Z_{eq.} = R + \frac{1}{i\omega C} + i\omega L$
 $= R + i\left(\frac{1}{\omega C} - \omega L\right) \Rightarrow \omega \text{ what } \omega \text{ is } |Z_{eq.}| \text{ smallest?}$

$\hookrightarrow \text{Im}[Z_{eq.}] = \omega L - \frac{1}{\omega C} = 0, \text{ when } \omega_0 = \omega = \frac{1}{\sqrt{LC}}$

$\omega_0 = \left[(2 \times 10^{-3} \text{ H})(2 \times 10^{-6} \text{ F}) \right]^{-\frac{1}{2}} = 5000 \sqrt{10} = 6.324 \times 10^4 \text{ rad/s}$

$f_0 = \frac{\omega}{2\pi} = \frac{5000\sqrt{10}}{2\pi} = 2.52 \times 10^3 \text{ Hz}$

b.) Quality Factor (Q) = $\frac{\omega_0}{\Delta\omega}$, where $\Delta\omega = \frac{R}{L}$: $Q = \frac{\omega_0}{(R/L)} = \frac{\omega_0 L}{R} = \frac{[(5000\sqrt{10})(2 \times 10^{-3})]}{3}$

c.) $Z = \sqrt{R^2 + (X_L - X_C)^2}$
 @ Resonance: $X_C = X_L$: $Z = \sqrt{R^2 + 0} = R, Z = R: 3 \Omega$

≈ 10.54

2) Design a RC low pass filter circuit that attenuates the voltage of a 1 kHz input signal by a factor of 10. (i.e. give the values of a resistor and capacitor that could be used to build such a circuit). What is the cutoff frequency for this circuit?

$$g = \frac{|\tilde{V}_{out}|}{|V_{in}|} = \frac{V_C}{V_{series}} = \frac{IX_C}{IZ_{series}} = \frac{1/\omega C}{\sqrt{R^2 + (1/\omega C)^2}} = \frac{1}{\sqrt{1 + \omega^2 R^2 C^2}} = \frac{1}{10} \Rightarrow 10 = \sqrt{1 + \omega^2 R^2 C^2}$$

$$\hookrightarrow 100 = 1 + \omega^2 R^2 C^2 \Rightarrow 99 = \omega^2 R^2 C^2 \Rightarrow \sqrt{99} = \omega RC$$

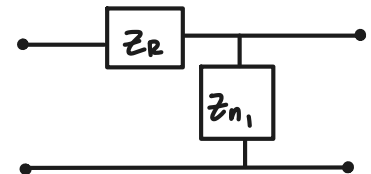
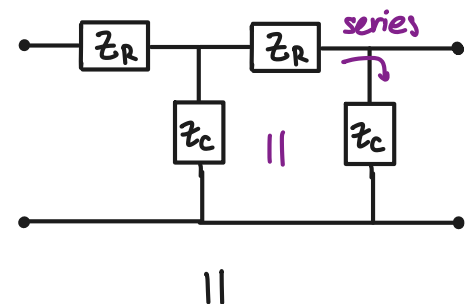
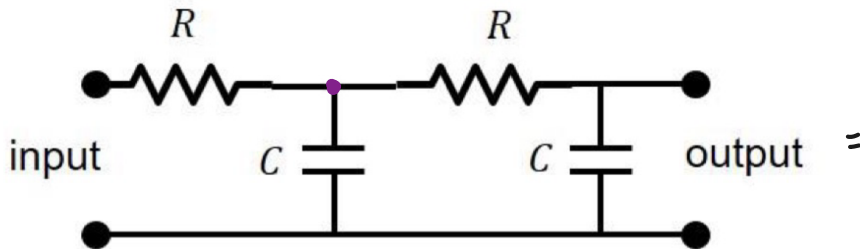
1KHz signal input

$$\Rightarrow \omega / \omega = 2\pi(1000) \Rightarrow \sqrt{99} = [2\pi(1000)]RC \Rightarrow \frac{\sqrt{99}}{2\pi(1000)} = RC \approx 1.58 \text{ ms}$$

$$\hookrightarrow R \cdot C = 1.58 \times 10^{-3}$$

if R & $C = 1.58 \text{ K}\Omega$ $1.0 \mu\text{F}$, then the cutoff frequency $\Rightarrow f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi(1.58 \times 10^{-3})} = 100.73 \text{ Hz}$

3) What happens if we take the output of a low-pass filter and send it through a second low-pass filter with the same R and C ? First compute the equivalent impedance of this circuit as seen from the input (assuming no load on the output). Then compute $\left|\frac{V_{out}}{V_{in}}\right|^2$ for this cascaded circuit of two successive low-pass filters. Then sketch a plot of $\left|\frac{V_{out}}{V_{in}}\right|^2$ (on a Log-Log scale) and also sketch a plot for the case of passing through a single low pass filter for comparison.



a.)

$$Z_{node 1} = Z_C \parallel (R + Z_C), \quad \omega / Z_C = \frac{1}{i\omega C}$$

so, $Z_{in1} = R + (Z_C \parallel R + Z_C)$

b.)

$$H(\omega) = \frac{V_{out}}{V_{in}} = \frac{1}{1 + 3\frac{(i\omega RC)^2}{x} + \frac{(i\omega RC)^4}{x^2}}$$

$$\text{so, } \left|\frac{V_{out}}{V_{in}}\right|^2 = \frac{1}{(1 - x^2)^2 + (3x)^2} = \frac{1}{1 + 7x^2 + x^4}$$

